



# Calculus/Tables of Derivatives

## General Rules

$$\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx}$$

$$\frac{d}{dx}(c \cdot f) = c \cdot \frac{df}{dx}$$

$$\frac{d}{dx}(f \cdot g) = f \cdot \frac{dg}{dx} + g \cdot \frac{df}{dx}$$

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{-f \cdot \frac{dg}{dx} + g \cdot \frac{df}{dx}}{g^2}$$

$$\frac{d}{dx}[f(g(x))] = \frac{df}{dg} \cdot \frac{dg}{dx} = f'(g(x)) \cdot g'(x)$$

$$\frac{d^n}{dx^n} f(x)g(x) = \sum_{i=0}^n \binom{n}{i} f^{(n-i)}(x)g^{(i)}(x)$$

$$\frac{d}{dx}\left(\frac{1}{f}\right) = -\frac{f'}{f^2}$$



Wikipedia has related information at **Differentiation rules** ([https://en.wikipedia.org/wiki/Differentiation\\_rules](https://en.wikipedia.org/wiki/Differentiation_rules)).

## Powers and Polynomials

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}x = 1$
- $\frac{d}{dx}x^n = nx^{n-1}$
- $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$
- $\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}$
- $\frac{d}{dx}(c_n x^n + c_{n-1} x^{n-1} + c_{n-2} x^{n-2} + \cdots + c_2 x^2 + c_1 x + c_0) = nc_n x^{n-1} + (n-1)c_{n-1} x^{n-2} + (n-2)c_{n-2} x^{n-3} + \cdots + 2c_2 x + c_1$

## Trigonometric Functions

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

## Exponential and Logarithmic Functions

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- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} a^x = a^x \ln(a) \quad \text{if } a > 0$
- $\frac{d}{dx} \ln(x) = \frac{1}{x}$
- $\frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)} \quad \text{if } a > 0, a \neq 1$
- $\frac{d}{dx} (f^g) = \frac{d}{dx} (e^{g \ln(f)}) = f^g \left( f' \frac{g}{f} + g' \ln(f) \right), \quad f > 0$
- $\frac{d}{dx} (c^f) = \frac{d}{dx} (e^{f \ln(c)}) = c^f \ln(c) \cdot f'$

## Inverse Trigonometric Functions

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$$\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arccos(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arctan(x) = \frac{1}{x^2 + 1}$$

$$\frac{d}{dx} \operatorname{arccot}(x) = -\frac{1}{x^2 + 1}$$

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{1}{|x| \sqrt{x^2 - 1}}$$

$$\frac{d}{dx} \operatorname{arccsc}(x) = -\frac{1}{|x| \sqrt{x^2 - 1}}$$

## Hyperbolic and Inverse Hyperbolic Functions

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$$\frac{d}{dx} \sinh(x) = \cosh(x)$$

$$\frac{d}{dx} \cosh(x) = \sinh(x)$$

$$\frac{d}{dx} \tanh(x) = \operatorname{sech}^2(x)$$

$$\frac{d}{dx} \operatorname{sech}(x) = -\tanh(x)\operatorname{sech}(x)$$

$$\frac{d}{dx} \coth(x) = -\operatorname{csch}^2(x)$$

$$\frac{d}{dx} \operatorname{csch}(x) = -\coth(x)\operatorname{csch}(x)$$

$$\frac{d}{dx} \operatorname{arsinh}(x) = \frac{1}{\sqrt{x^2 + 1}}$$

$$\frac{d}{dx} \operatorname{arcosh}(x) = \frac{1}{\sqrt{x^2 - 1}}, \quad x > 1$$

$$\frac{d}{dx} \operatorname{artanh}(x) = \frac{1}{1 - x^2}, \quad |x| < 1$$

$$\frac{d}{dx} \operatorname{arsech}(x) = -\frac{1}{x\sqrt{1 - x^2}}, \quad 0 < x < 1$$

$$\frac{d}{dx} \operatorname{arcoth}(x) = \frac{1}{1 - x^2}, \quad |x| > 1$$

$$\frac{d}{dx} \operatorname{arcsch}(x) = -\frac{1}{|x|\sqrt{1 + x^2}}, \quad x \neq 0$$

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